

CONSISTENCY OF THE k_n -NEAREST NEIGHBOR RULE UNDER ADAPTIVE SAMPLING

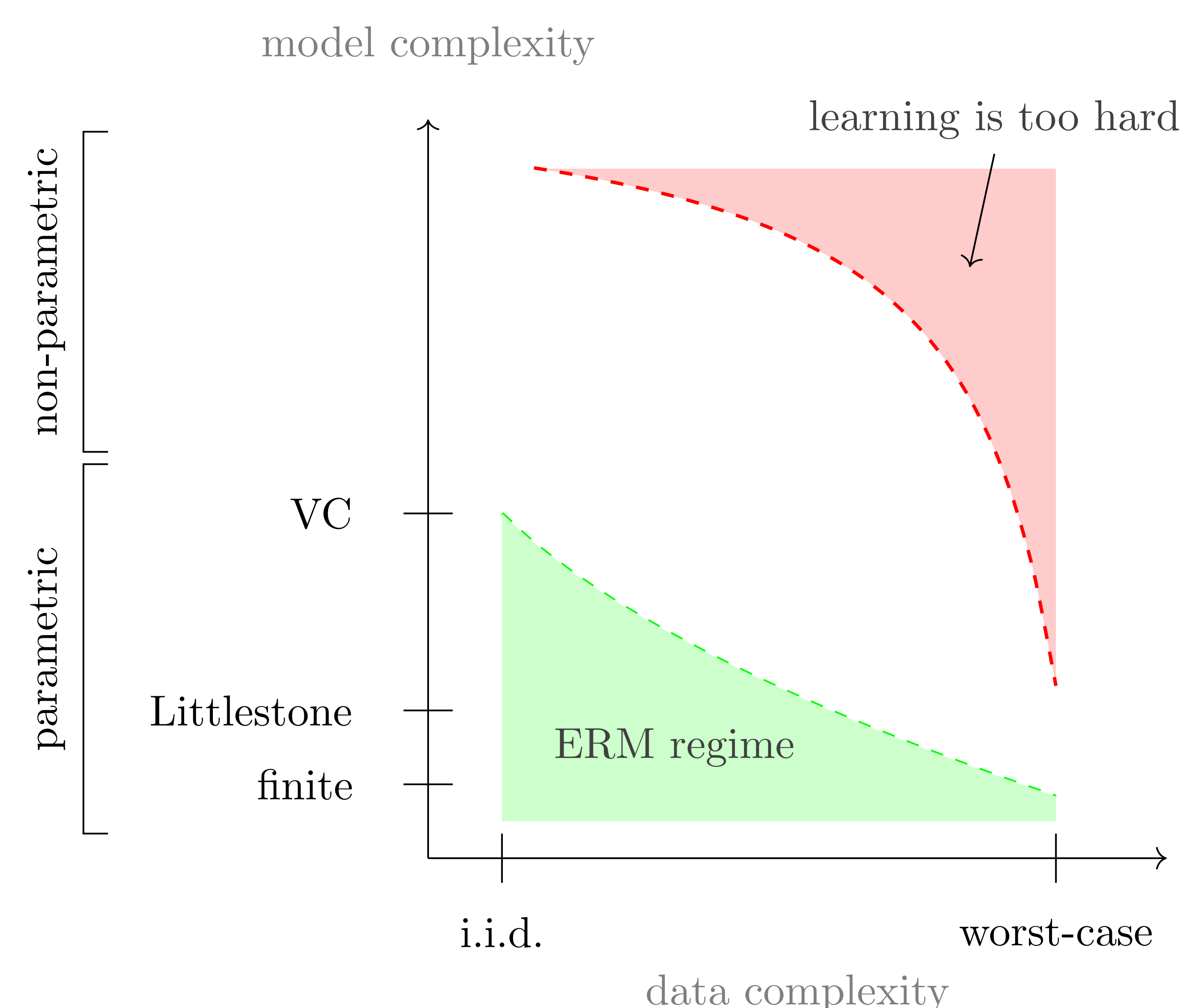
Robi Bhattacharjee¹, Sanjoy Dasgupta² and Geelon So²

¹University of Tübingen and Tübingen AI Center, ²University of California, San Diego

MOTIVATION

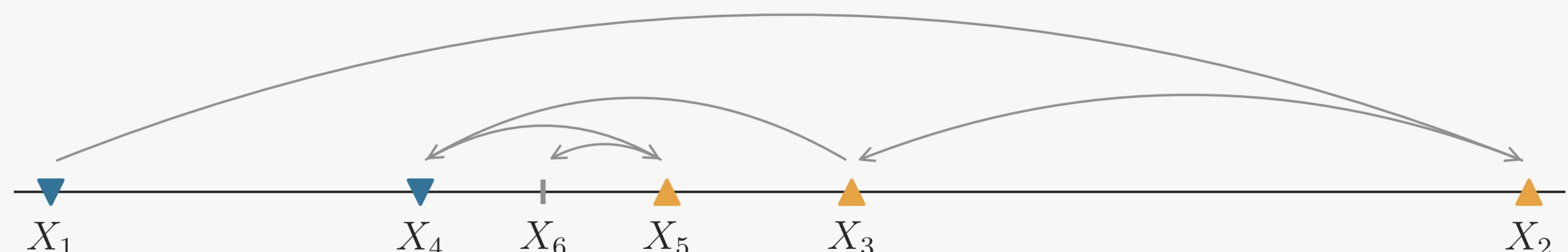
Question: can we learn from adaptive data?

- *Scientific process*—past data influences what experiment is performed next.
- *Exploratory/adaptive data analysis*—the question asked may depend on available data.
- *Online decision making*—data improves business operations, which affects further data.



TAKEAWAYS

- Adaptivity can generate **patterns out of noise**.
- Worst-case adaptive data extremely pathological.
- **Smoothed analysis** gets at more “realistic” data.
- The k_n -nearest neighbor rule is consistent under very mild statistical and structural assumptions.



Making patterns out of noise. Let $\mathcal{X} = \mathbb{R}$ and let $\eta \equiv 1/2$ everywhere (labels are generated by unbiased coin flips). If the instances X_n are sampled using the binary search procedure, this generates a data set that looks linearly separable.

LEARNING PROBLEM

Binary classification with noise

- Data space: $\mathcal{X} \times \mathcal{Y}$ where $\mathcal{Y} = \{0,1\}$
- Conditional label mean: $\eta : \mathcal{X} \rightarrow [0,1]$

$$\eta(x) \equiv \Pr(Y = 1 \mid X = x)$$
- Metric and measure: ρ and ν

ADAPTIVE SAMPLING

For all time $n = 1, 2, \dots$

- Adversary selects a test distribution μ_n
- Test instance $X_n \sim \mu_n$ is drawn
- Learner predicts label $\hat{Y}_n$
- Nature reveals correct label $Y_n \sim \text{Ber}(\eta(X_n))$

LEARNING GOAL: CONSISTENCY

Consistency with Bayes classifier:

$$\limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \mathbf{1}\{\hat{Y}_n \neq Y_n^*\} = 0 \quad \text{a.s.}$$

where Y_n^* is the Bayes-optimal label for X_n .

THE k_n -NEAREST NEIGHBOR RULE

Prediction rule for X_n :

- Find the k_n closest data points to X_n .
- Return the majority vote over their labels.

SMOOTHED DATA PROCESSES

Definition. A data process $\{X_n, Y_n\}_{n=1}^\infty$ is *uniformly dominated* ν if for all $\varepsilon > 0$, there is a $\delta > 0$ such that when a measurable set A satisfies $\nu(A) < \delta$, then:

$$\Pr(X_n \in A \mid X_{1:n-1}, Y_{1:n-1}) < \varepsilon.$$

It is *Lipschitz dominated* or *L-smoothed* if $\varepsilon(\delta) \leq L\delta$.

MAIN RESULT

Assumptions.

- Let $(\mathcal{X}, \rho, \nu)$ be an upper doubling space.
- Let $(k_n)_n$ grow between $\omega(\log n)$ and $o(n/\log n)$ in a fairly smooth way.
- Let η be measurable.

Theorem. If the data process is Lipschitz dominated, then the k_n -nearest neighbor rule is consistent.